

If only you could find that video where... well, multimedia search is only a question of when, not how

Image Ready

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We shall not begin this with “a picture is worth a thousand words.” Sorry, we just did. Well, whatever we start with, the relevance of that adage here is threefold: first, we mostly search for text; second, 99 per cent of the actual data on the Web is audio-visual (AV); and third, we’re going to be searching for non-textual information more and more. Witness the exploding popularity of online video, for example.

The Big Picture

The problem is simple enough: search engines do the text spidering, and all that remains in that domain is how to produce more relevant results. Even if we still mostly search for text, the fundamental problem there—or its solution—won’t change. Now when a search engine like Google or Yahoo! gives you image results, it’s based on metadata and tags: *someone* has to have spelt out in some way or the other that a certain image depicts Britney Spears (with her hair in place) and not her grandmother. What when such information isn’t available? What when you pose in front of the Taj Mahal holding a cigarette, and the image is tagged “Taj Mahal”: would you find it by searching for yourself—or for “cigarette”? What when Bill Gates meets up with our IT minister, and you want a picture of the minister? There is, quite literally, an infinity of such questions. Fact is, there’s tons of AV content out there, and only things like tags are used to search it—at least in the case of the big search engines.

Imaging Pravin Warhokar
Photograph Amrut Patki

Looking At The Words

February 8, 2007, it was announced that the European Commission had granted funding to a research project that would be led by a search technology firm called Fast Search and Transfer (FAST). Without going into the specifics, FAST will now be working to innovate search solutions across all types of content, including multimedia.

From a FAST white paper, roughly, standard search is built on four pillars: “treating a load of information as discrete documents; using a distributed architecture for optimal performance; taking free text and metadata to derive each document’s ‘meaning’; and using a relevancy model to best match each query to the information in the index.

“Multimedia search faces basic challenges to this paradigm: ‘documents’ are not clearly defined; particularly large data volumes are involved; what each object talks about is not simple to extract; and relevancy models need to take more variables into account.”

Let’s take a look at each of these challenges.

Looking At The Picture

First, “documents” are not clearly defined. A single news programme can be divided into many parts, with each reportage clip as a separate part. Second, there’s a lot of AV information out there, in comparison to text: only 1 per cent of information—in terms of data—is text, by some estimates. Third and most important, knowledge extraction from text is *relatively* simple. Yes, it’s not simple at all, but *relatively*, it is. After all, a machine can be trained to classify stuff with words like “stocks” and “shares” as financial, but

